PHASE 05 / PROJECT DOCUMENTATION

Project Development

Preparing EV data and implementing the planned analysis experience.

This document records the implementation workflow from source inspection through preprocessing, metric creation, visualization, and result reporting.

01 | Implementation approach

Development begins with the real dataset rather than assumptions about what it contains. The team should preserve the original source, prepare a separate analysis copy, and keep a short record of each transformation. Dashboard features must reflect available fields and validated calculations.

02 | Dataset specification record

Item	Complete with actual project details
Dataset title and publisher	[Enter the exact title and source]
Source URL and access date	[Enter URL and date accessed]
License / redistribution terms	[Enter the applicable terms]
File type and version	[CSV / Excel / other; filename and date]
Rows, columns, and coverage	[Record after loading and profiling]
Relevant fields and units	[Use actual column names and units]
Known gaps	[Missing fields, time coverage, sampling limitations]

03 | Data preparation workflow

1. Create a working copy and retain the original source unchanged.
2. Inspect schema, data types, row counts, unique values, and summary statistics.
3. Check duplicates and define duplicate rules before removing any records.
4. Review missing values field by field; keep unknown values distinct from genuine zero values.
5. Standardize manufacturer, category, battery, and country labels without losing meaningful distinctions.
6. Convert numeric fields safely and investigate invalid or implausible values.
7. Normalize units such as kilometres/miles and clearly record any conversions.
8. Check whether time fields and sales values are comparable across records.
9. Calculate derived measures only when the required inputs and denominator are valid.
10. Recheck counts and totals, then document all material cleaning decisions.

04 | Business question to analysis

Question	Potential output	Validation before use
Which categories have the highest recorded sales?	Sorted bars or yearly trend	Confirm sales units, period, and coverage
How do manufacturers compare?	Ranking or share chart	Define comparison scope and valid denominator
What battery types are represented?	Distribution chart	Normalize battery labels and preserve unknowns
How does range vary?	Category comparison or distribution plot	Use consistent units and consider sample size
Which production locations appear most often?	Country chart/map	Verify what the location field actually means
How do ratings differ?	Rating distribution/category chart	Confirm scale and treatment of missing ratings

05 | Build checklist

- Create a clear overview and avoid KPIs that cannot be supported by the dataset.
- Use chart titles that state the question or measure, with units and period.
- Add only filters connected to actual fields and test combinations.
- Document calculation logic for sales, averages, rankings, and shares.
- Add source and limitation notes to the dashboard or repository.
- Capture screenshots after the dashboard is stable and tested.
- If publishing online, verify the access settings and test the public URL.

06 | Results log template

Question / chart	Observed result from data	Interpretation and caveat
[Question]	[Verified value or pattern]	[Careful meaning; scope/limitation]
[Question]	[Verified value or pattern]	[Careful meaning; scope/limitation]
[Question]	[Verified value or pattern]	[Careful meaning; scope/limitation]

Development note

Do not state a company, battery chemistry, country, or category as a market leader until the claim is calculated from the selected dataset and qualified by its coverage. The dashboard should show what the data supports, not imply more than it can establish.